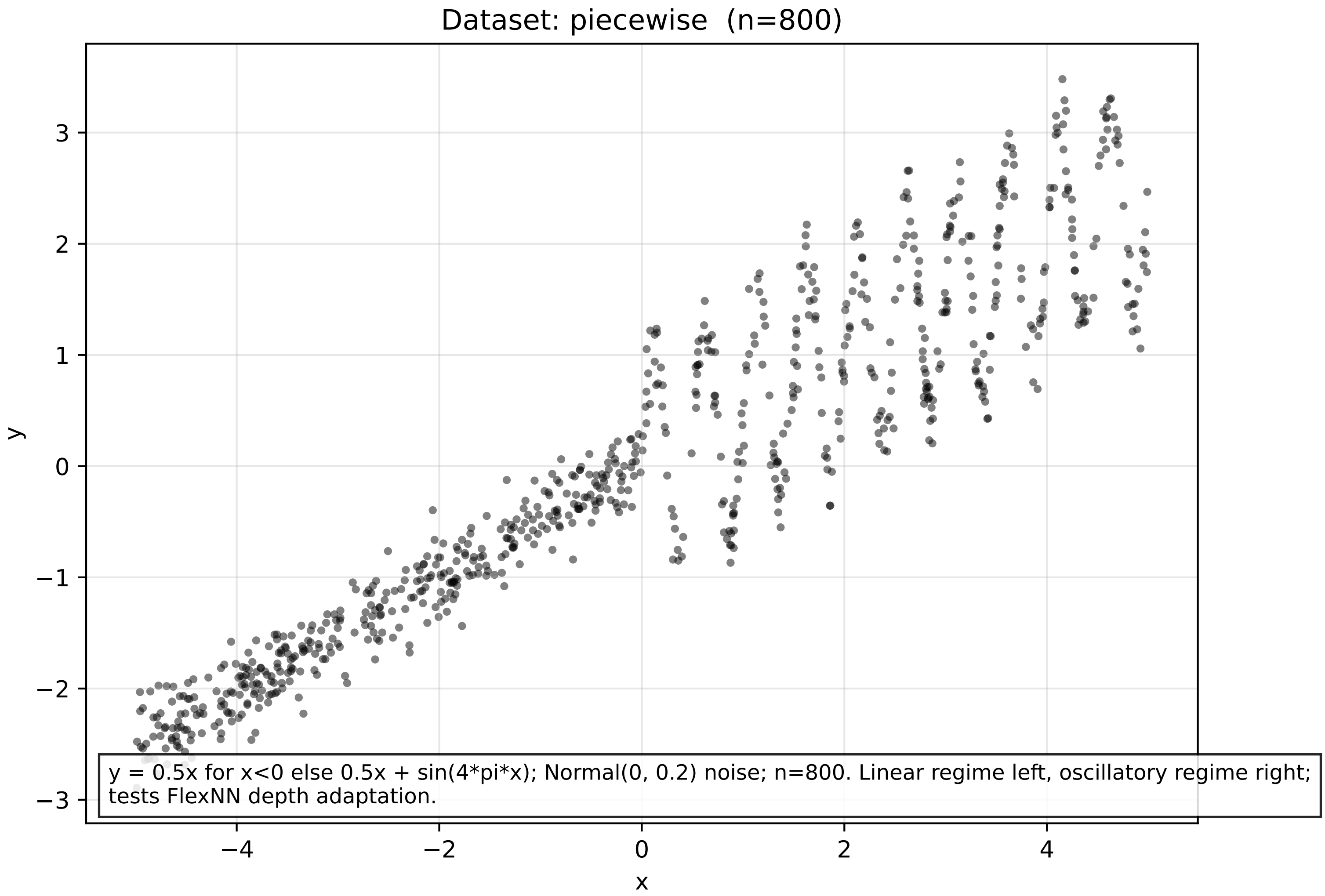
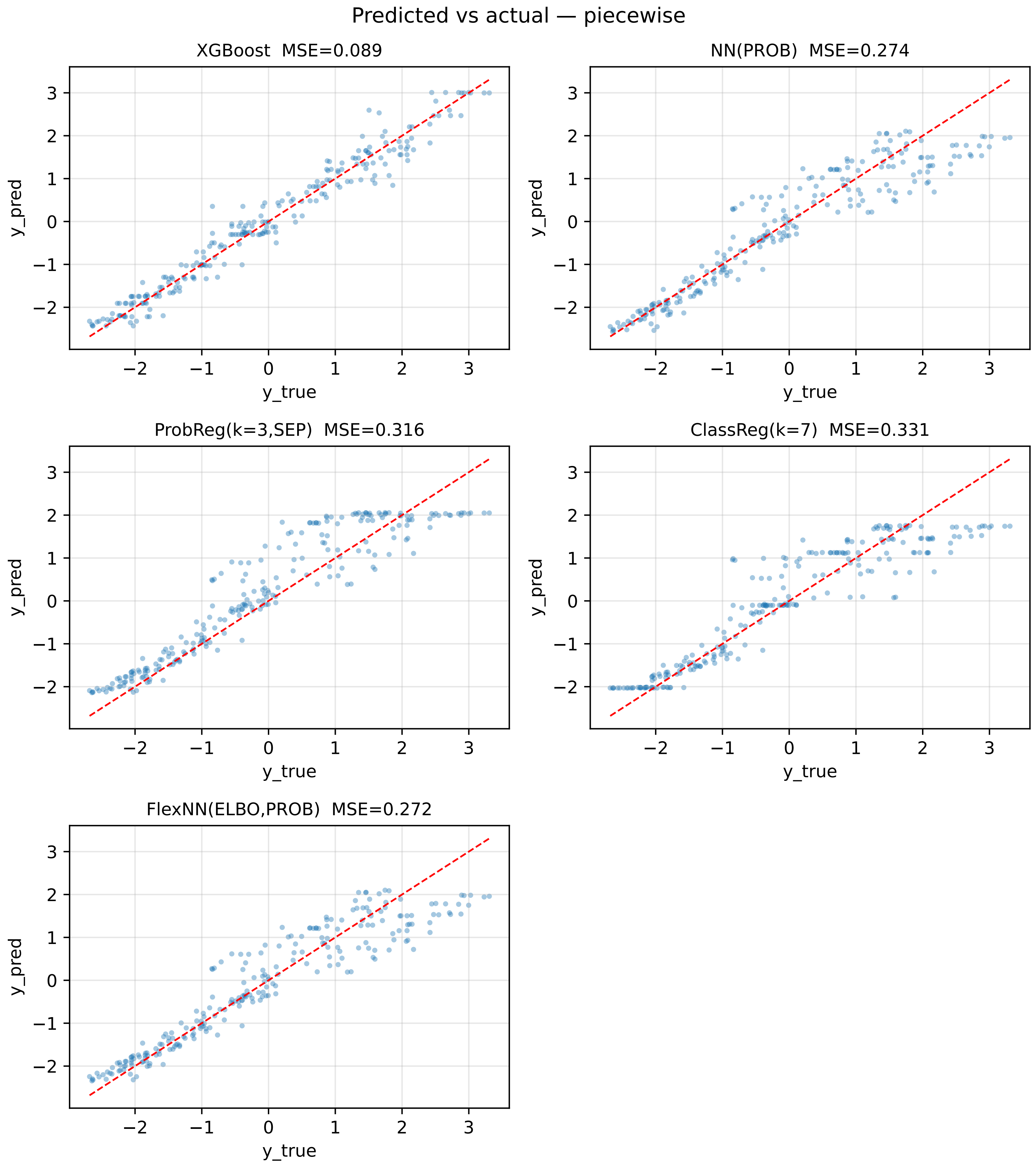
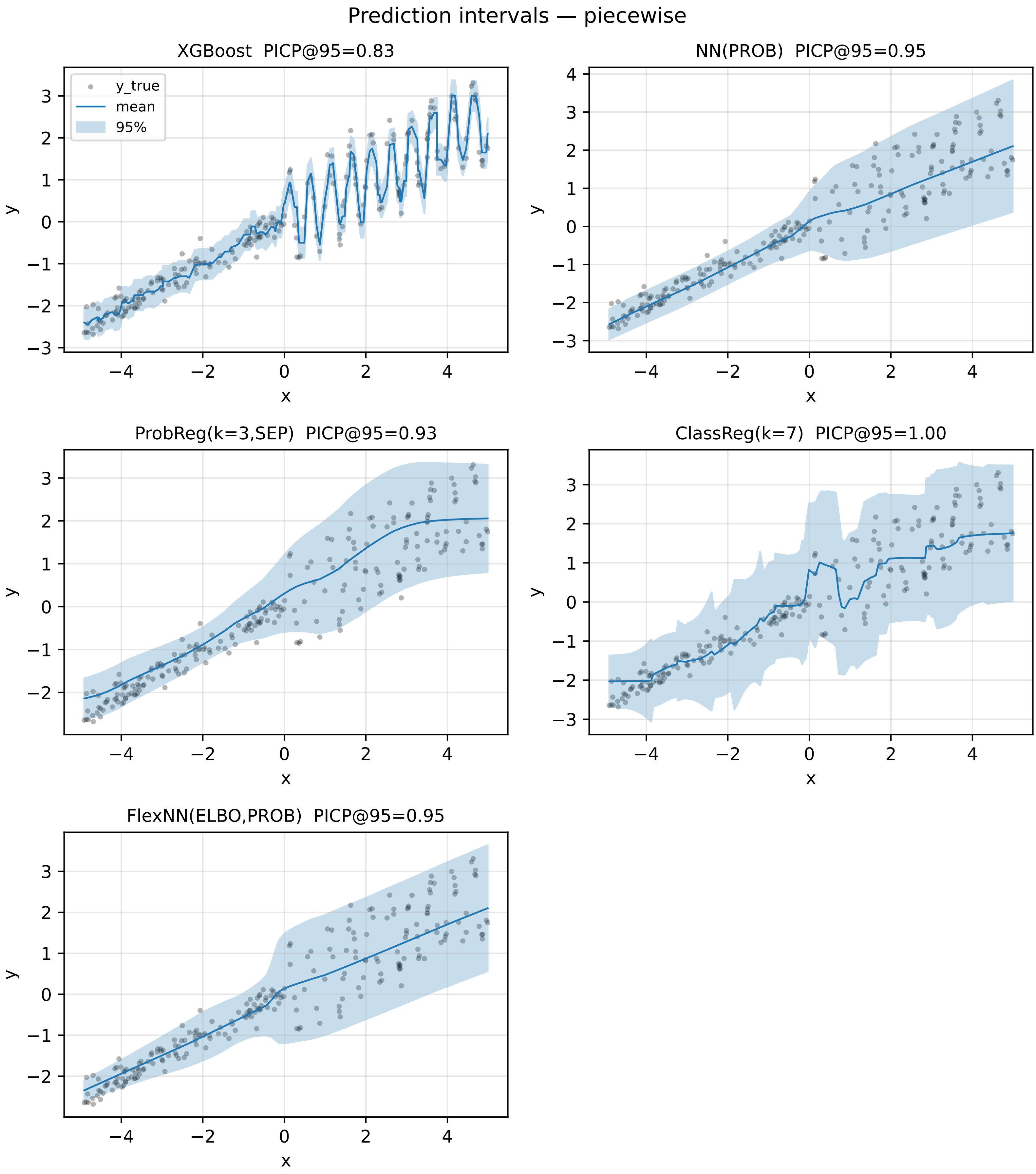

Dataset: piecewise (n=800)
y
x
3
2
1
0
−1
−2
−3
−4
−2
0
2
4
y = 0.5x for x<0 else 0.5x + sin(4*pi*x); Normal(0, 0.2) noise; n=800. Linear regime left, oscillatory regime right; tests FlexNN depth adaptation.

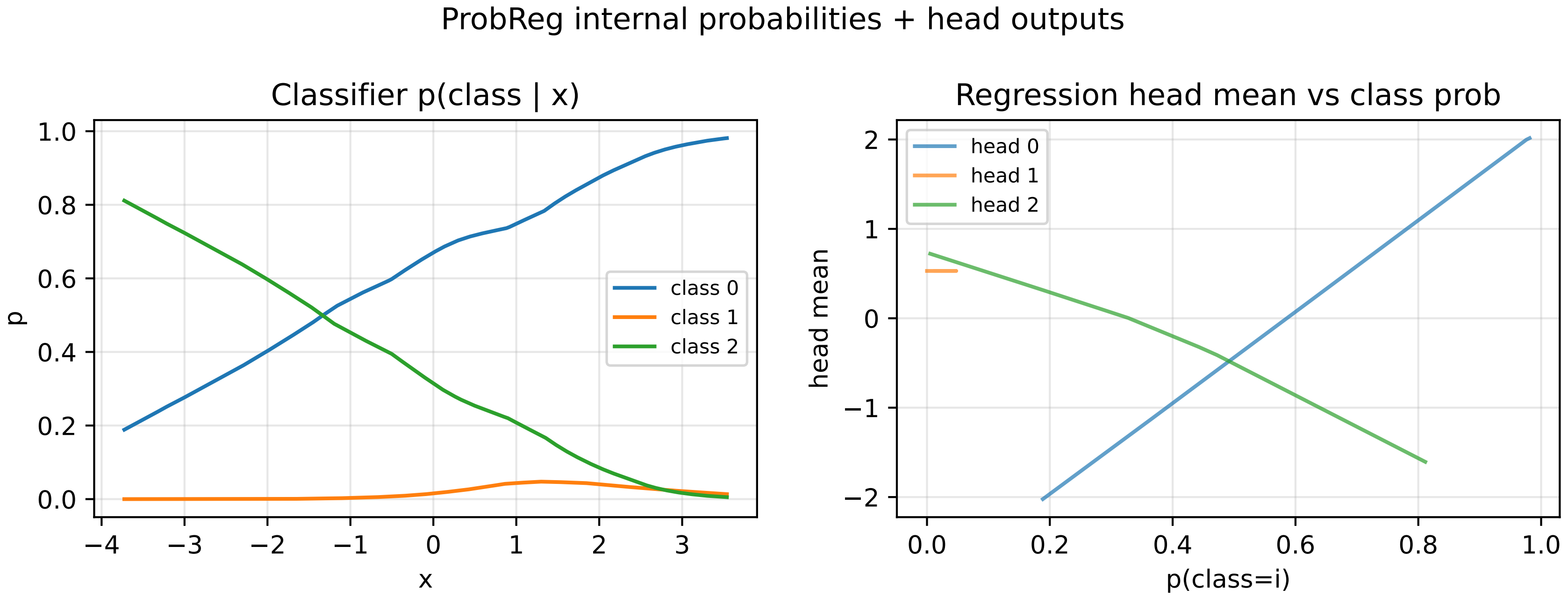
Model comparison

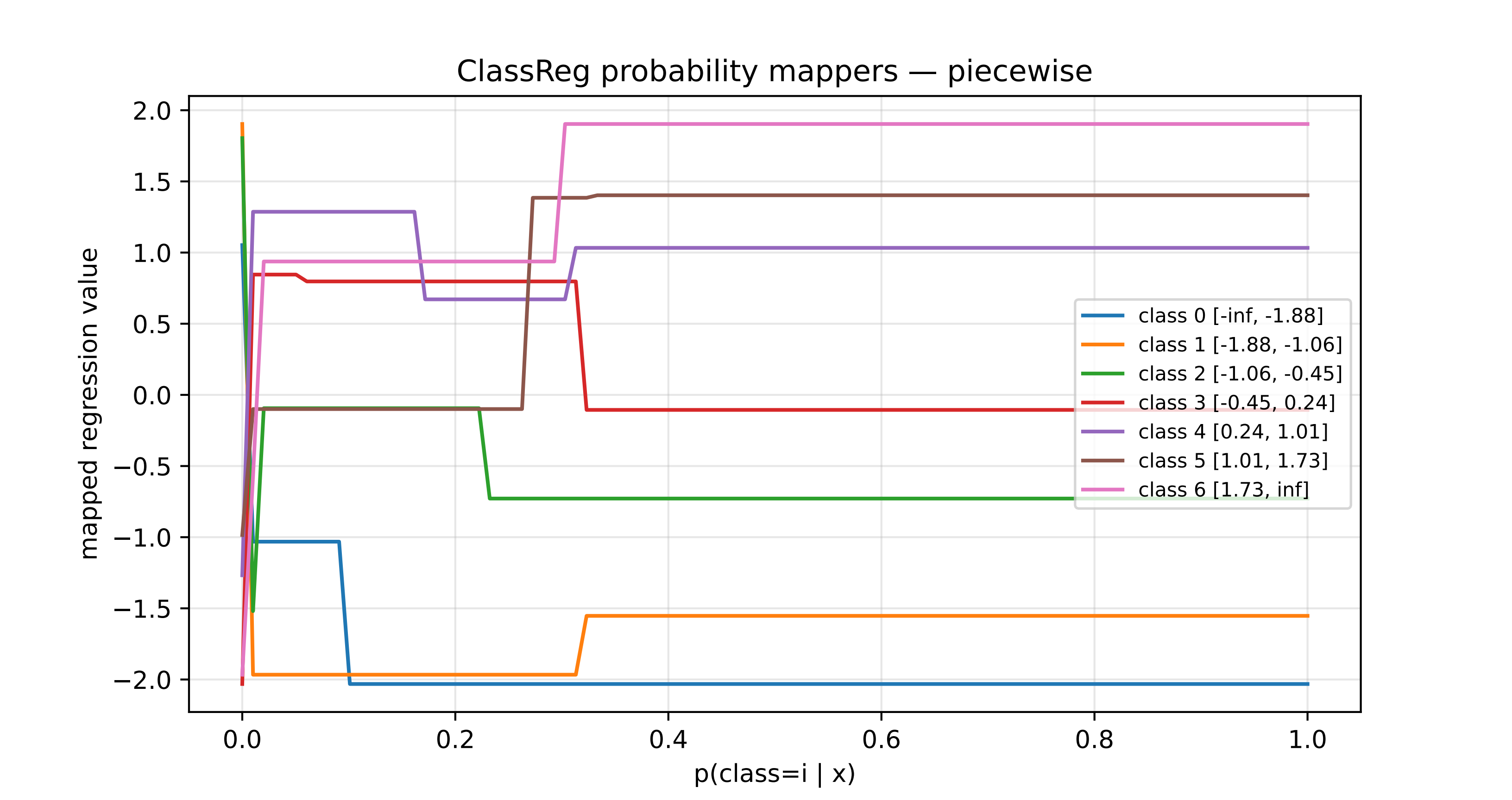
model	mse	nll	crps	picp95	mpiw95	sharpness
XGBoost	0.0889	0.5239	0.1686	0.8292	0.7281	0.1857
NN(PROB)	0.2735	0.4797	0.2631	0.9542	1.8734	0.4779
ProbReg(k=3,	0.3164	0.6563	0.2976	0.9292	1.8002	0.4593
ClassReg(k=7	0.3309	0.8143	0.3104	1.0000	2.8493	0.7269
FlexNN(ELBO,	0.2723	0.5501	0.2667	0.9500	1.9327	0.4930



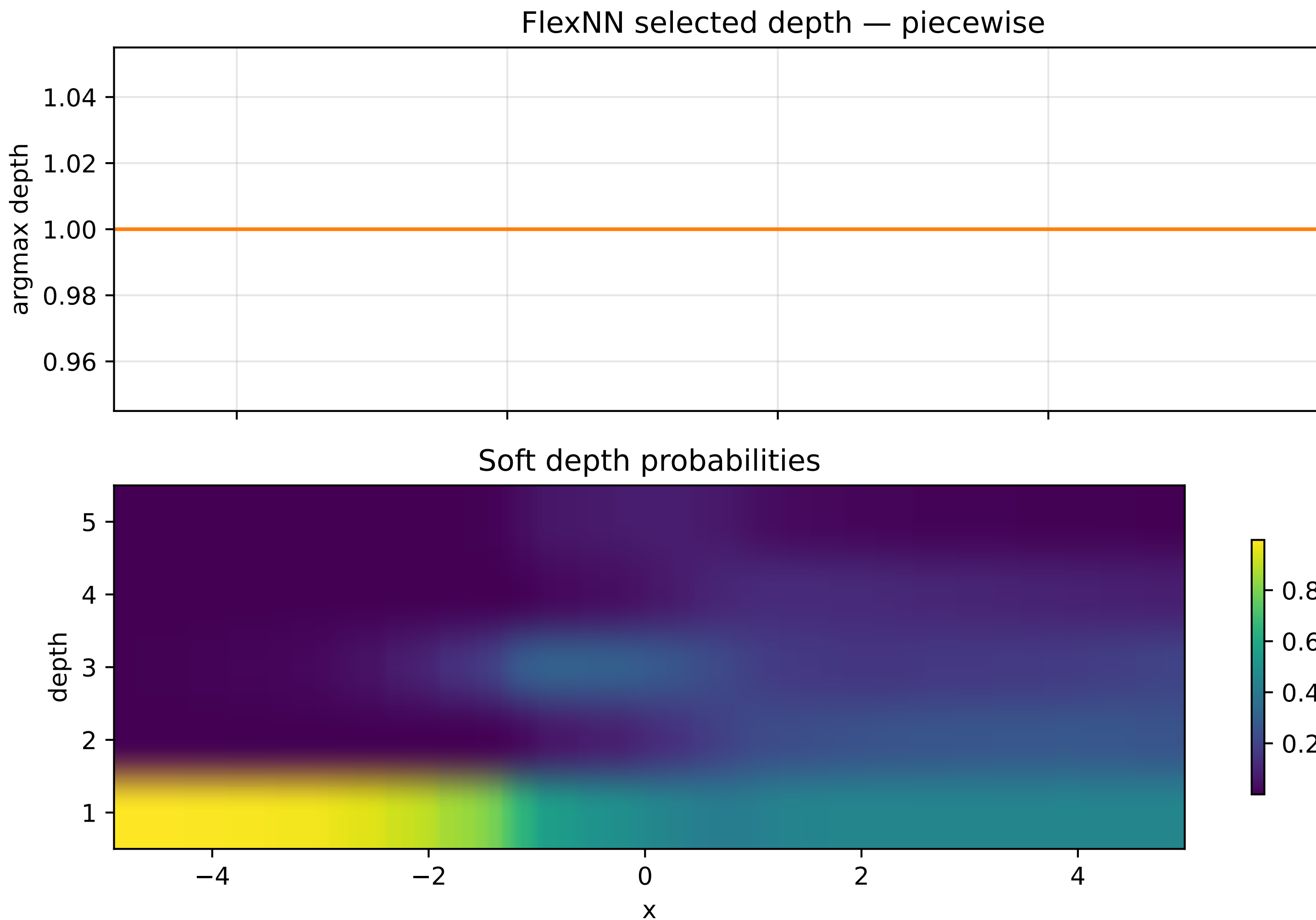
Predicted vs actual — piecewise
XGBoost MSE=0.089
NN(PROB) MSE=0.274
ProbReg(k=3,SEP) MSE=0.316
ClassReg(k=7) MSE=0.331
FlexNN(ELBO,PROB) MSE=0.272
y_pred
y_true
−2
−1
0
1
2
3


Prediction intervals — piecewise
XGBoost PICP@95=0.83
y_true
mean
95%
NN(PROB) PICP@95=0.95
ProbReg(k=3,SEP) PICP@95=0.93
ClassReg(k=7) PICP@95=1.00
FlexNN(ELBO,PROB) PICP@95=0.95
x
y


ProbReg internal probabilities + head outputs
Classifier p(class | x)
p
x
class 0
class 1
class 2
1.0
0.8
0.6
0.4
0.2
0.0
−4
−3
−2
−1
0
1
2
3
Regression head mean vs class prob
head mean
p(class=i)
head 0
head 1
head 2
2
1
0
−1
−2
0.0
0.2
0.4
0.6
0.8
1.0



ClassReg probability mappers — piecewise
mapped regression value
p(class=i | x)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0.0
0.2
0.4
0.6
0.8
1.0
class 0 [-inf, -1.88]
class 1 [-1.88, -1.06]
class 2 [-1.06, -0.45]
class 3 [-0.45, 0.24]
class 4 [0.24, 1.01]
class 5 [1.01, 1.73]
class 6 [1.73, inf]



FlexNN selected depth — piecewise
argmax depth
1.04
1.02
1.00
0.98
0.96
Soft depth probabilities
depth
5
4
3
2
1
−4
−2
0
2
4
x
0.8
0.6
0.4
0.2

Ranked MSE for piecewise:
- XGBoost: MSE=0.0889 NLL=0.524 PICP95=0.83
- FlexNN(ELBO,PROB): MSE=0.2723 NLL=0.550 PICP95=0.95
- NN(PROB): MSE=0.2735 NLL=0.480 PICP95=0.95
- ProbReg(k=3,SEP): MSE=0.3164 NLL=0.656 PICP95=0.93
- ClassReg(k=7): MSE=0.3309 NLL=0.814 PICP95=1.00

Notes:
Piecewise linear/oscillatory: FlexNN should use shallow depth in the linear regime (x<0) and deep